

Hamilton Ferris

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Education

Winona State University

BACHELOR OF SCIENCE - COMPUTER SCIENCE

Winona, MN

August 2022 - May 2026

Relevant Coursework

OPERATING SYSTEMS

August 2026 - December 2026

Study of the principles and design techniques of operating systems. Topics include concurrent processes, scheduling, deadlocks, memory management, file and directory organizations and protection/security.

COMPUTER SYSTEMS

January 2025 - May 2025

Topics include computer organization and computer architecture, data representation, assembly language, memory systems, operating systems, networking and security, run-time environments, and advanced topics such as RISC vs. CISC, non von Neumann architectures, and Java virtual machine.

DATA STRUCTURES

January 2025 - May 2025

A detailed study of more advanced data structures and algorithms, including concepts and techniques of design efficiency and complexity of algorithms and their lower bounds.

Experience

Lead Student Systems Administrator

Winona, MN

WINONA STATE UNIVERSITY

2023 - Present

- Created, provisioned, and maintained Linux virtual machines, optimizing server performance and reducing Windows server load significantly.
- Mentored four student workers, boosting administration success and retention by 1.5 years.
- Automated certificate renewal processes, decreasing worst-case time to completion by up to 850%.
- Deployed firewall and proxy servers, enhancing network efficiency by freeing up to 20% of bandwidth.
- Implemented security fixes to strengthen system integrity and eliminate vulnerabilities.

Projects

Odisk

DESIGNER/DEVELOPER

June 2025 - Present

- Used the inotify library to monitor filesystem read/write events and format the output for internal API consumption.
- Built a custom INI parser to gain deeper insight into file parsing mechanics and enhance control over configuration handling.

Automatic Certificate Renewal

DESIGNER/DEVELOPER

November 2024 - January 2025

- Automated Let's Encrypt certificate renewal utilizing the Azure-DNS plugin for seamless domain validation, allowing multiple redundant scripts to be deleted.
- Optimized certificate renewal script with bash scripting, reducing total execution time by up to 850%.

SCVMM Checkpoint API

DEVELOPER

February 2023 - May 2023

- Developed and integrated an internal SCVMM API to automate the creation and deletion of virtual machine checkpoints.
- Built a JSON API service using Flask, Waitress, and Python typing libraries for structured request handling.
- Implemented a PowerShell script to convert Python functions into SCVMM-compatible automation tasks.

My Resume

DESIGNER/DEVELOPER

July 2023 - Present

- Designed a fully customized resume using Typst libraries, incorporating personalized formatting and layout for professional presentation.
- Automated resume deployment by configuring GitHub Actions to generate and publish the latest PDF version to a designated release.